

Full-Stack Developer

Education

Alamein International University (AIU) — Alexandria, Egypt

Expected May 2027

Bachelor of Science (B.Sc.), Computer Science

- **Coursework:** Data Structures and Algorithms, System Design, Advanced Database Systems, Information Security, Object-Oriented Programming

Work Experience

Full-Stack Developer

Jan 2025 – Present

Paradise Sharm Tours

paradisesharm.com

- Architected and deployed paradisesharm.com, a multi-language (EN/AR/RU/DE/IT) tour booking platform using Next.js, Express.js, MongoDB, and Traefik with automated SSL/TLS, serving 174 MAU across 6 countries (US, EG, IT, IE, SE, UK)
- Built observability stack with Prometheus, Grafana, and k6 stress testing (up to 220 concurrent VUs, 2-hour soak tests) enabling proactive performance regression detection before deployment
- Integrated Google Analytics, PostHog, and Plerdy across the platform, tracking behavioral funnels for 651 total users and surfacing conversion drop-off insights that informed two UX iterations
- Optimized frontend to achieve 91 Performance, 100 Accessibility, and 92 Best Practices Lighthouse scores with 0.9s FCP and 0 cumulative layout shift
- Hardened application security with RBAC, JWT refresh token rotation, and rate limiting; implemented automated daily MongoDB backups with 7-day retention policy

Projects

Cairo Transportation Network | C#, .NET, Next.js, SQLite, EF Core

[Live Demo](#)

- Engineered a smart-city routing platform processing 35 locations and 74 roads across Greater Cairo using A* emergency routing (39% faster than Dijkstra, 40% fewer node expansions) and 0/1 Knapsack DP for maintenance budget optimization (42% improvement over greedy, serving 78.5% transit demand with 50 vehicles)
- Built a modular monolith ASP.NET Core REST API with IMemoryCache (< 1ms graph retrieval), Next.js interactive Leaflet map, and automated CI/CD via GitHub Actions, achieving < 2ms average algorithm response time

E-commerce API | TypeScript, Node.js, Express, MongoDB, Stripe

[GitHub](#)

- Built a production-ready ecommerce REST API with 8+ modules (products, orders, cart, reviews, coupons) featuring Stripe payment integration with webhook handling and email verification flows
- Implemented BaseController pattern for DRY CRUD operations, JWT authentication with MFA-protected account deletion, and hardened security (Helmet, NoSQL injection protection, HPP, rate limiting, input sanitization)

Technical Skills

Languages: TypeScript, C#, JavaScript, Java, Python, SQL, HTML, CSS

Frontend: Next.js, React, React Query, TailwindCSS, Framer Motion, Radix UI

Backend: Node.js, Express.js, .NET, ASP.NET, REST APIs

Databases: MongoDB, MariaDB, MySQL, PostgreSQL

DevOps: Docker, Docker Compose, Traefik, Cloudflare tunnel, Portainer, pnpm monorepo, Bash

Monitoring: Prometheus, Grafana, k6, Google Analytics, PostHog, Plerdy

Quality: Jest, Supertest, SonarQube, Husky, ESLint, Prettier

Certifications

Object-Oriented Programming Using Java — ITI (Information Technology Institute) — 24 training hours, project-based with HackerRank assessment (Jul-Aug 2025)

Leadership

Technical Head

Apr 2026 – Present

AIU ICPC Community — Alamein International University (AIU)

- Architecting and developing a full-stack competitive programming platform for a 600-member ICPC community using Next.js and Node.js, featuring problem archives, live contests with real-time leaderboards, and user rating profiles